

# Supplementary Material

ChiralFold: systematic detection of D-amino acid stereochemistry errors in the Protein Data Bank

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## 1 Supplementary Methods

### 1.1 Ramachandran benchmark protocol

Era-representative stratified sampling across X-ray ultra ( $\leq 1.20$  Å), high, medium, low, NMR, and cryo-EM bins (seed 20260612). mmCIF→PDB conversion uses chain-ID remapping (A–Z, 0–9). Structures with >36 chains skipped. Script: `benchmarks/expand_ramachandran_benchmark.py`.

### 1.2 Mirror-image, AF3 correction, and enumeration

$L \leftrightarrow D$  reflection:  $(x, y, z) \rightarrow (-x, y, z)$ . AF3 correction reflects violating  $C\alpha$  across the N–C– $C\beta$  plane. Diastereomer enumeration scores  $2^k$  L/D patterns. Interface scoring uses BSA, H-bonds, and salt bridges.

### 1.3 AF3-mimetic correction resource benchmark

Because AF3 outputs are not redistributed or rerun, we benchmarked synthetic AF3-mimetic Childs-system peptide PDBs with the published 51% AF3 D-peptide chirality violation rate as the external reference. Across DP19, DP9, and DP12 (39 auditable residues; 20 synthetic inverted stereocenters), ChiralFold detected 100% of inverted stereocenters, corrected 100% to the expected signed-volume class, and left 0% residual violations. The full `correct_af3_output` pipeline took 0.0367 s total across the three systems (mean 0.0122 s/system) with peak traced allocations below 0.14 MB; process RSS was dominated by the already-loaded Python/RDKit runtime. Data: `results/af3_resource_benchmark.json`.

### 1.4 Lean 4 formalization

Proofs in `formal/chirality_nogo/` (Lean 4.28.0, Mathlib). Build: `lake build`.

## 2 Supplementary Tables

### 2.1 Table S1: All 29 D-label/L-coordinate mismatches

See full machine-readable dataset `results/d_residue_verification.csv` and Tables 1–2 in the repository manuscript `paper/chiralfold_paper.tex`.

| Theorem                                               | File                  | Consequence                                  |
|-------------------------------------------------------|-----------------------|----------------------------------------------|
| same_distances_opposite_orientations_standardTet      | ChiralityNoGo         | Equal distances, opposite orient             |
| orientTarget_not_factorsThroughDistMatrix             | ModelAbstraction      | Volume cannot factor through d               |
| distance_data_determines_magnitude_not_chirality_sign | DistancesOrientVolume | Distances fix $ \text{orient} ^2$ , not sign |

## 2.2 Table S2: Formal no-go theorems

## 3 Supplementary Figures

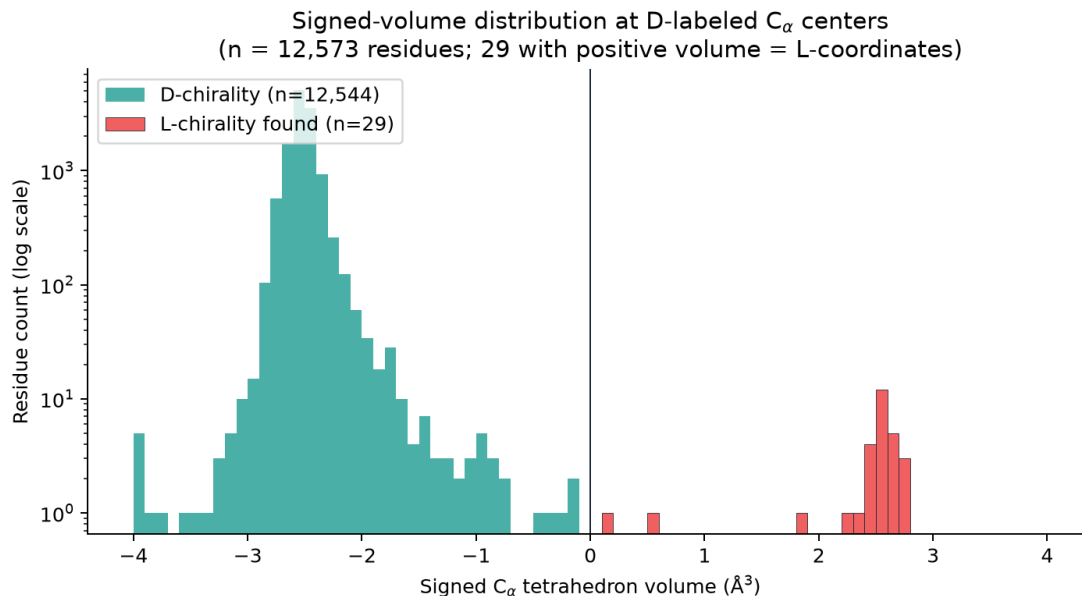


Figure 1: Signed volume distribution across 12,573 D-residues.

## 4 Extended statistics

Ramachandran  $n = 362$  (post-5M2K exclusion): `results/ramachandran_279struct_chainfix_summary.json`  
— Spearman  $\rho = 0.52$ ; Pearson  $r = 0.85$  (0.33 if 5M2K retained). Pilot  $n = 31$  Bland–Altman and bootstrap CIs: `paper/stats_extended.json` via `paper/stats_analysis.py` (seed 42). Deposition-year logistic OR= 1.60/year; CCD clustering  $\chi^2 = 113.78$  ( $p = 2.33 \times 10^{-16}$ ; nine codes with errors vs nine clean).

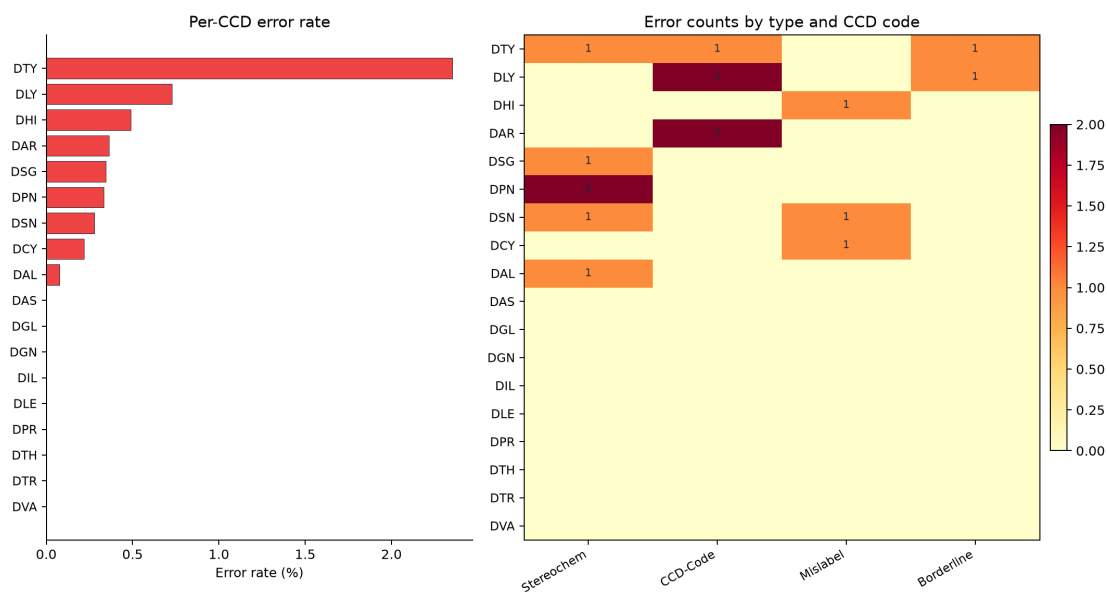


Figure 2: CCD code error clustering heatmap.

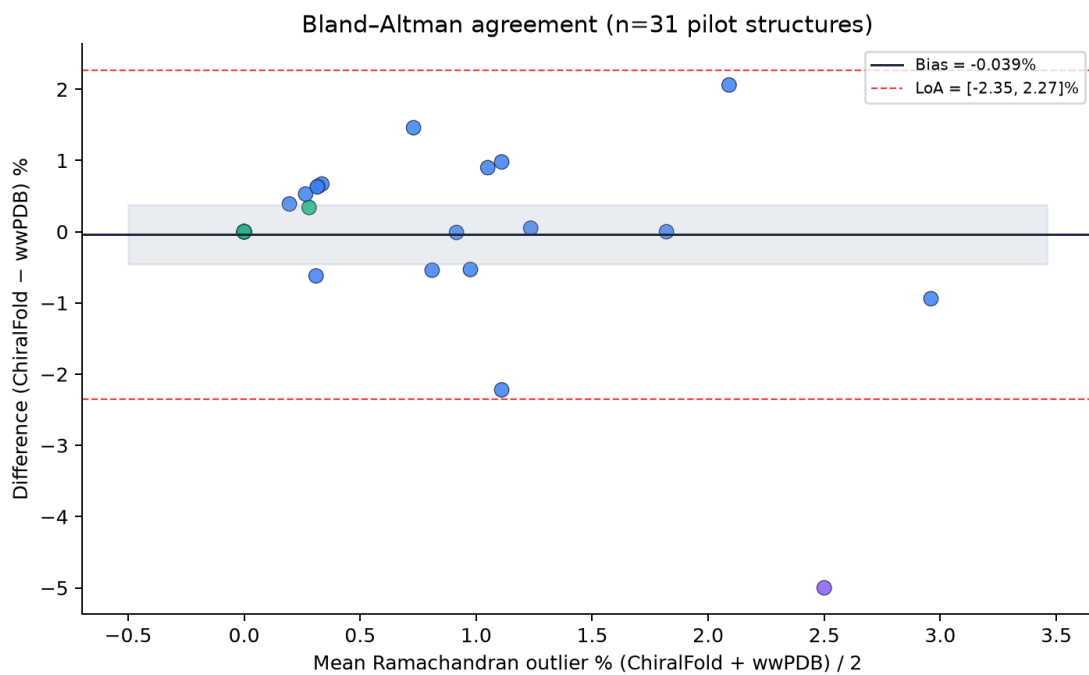


Figure 3: Bland-Altman plot, 31-structure pilot benchmark.

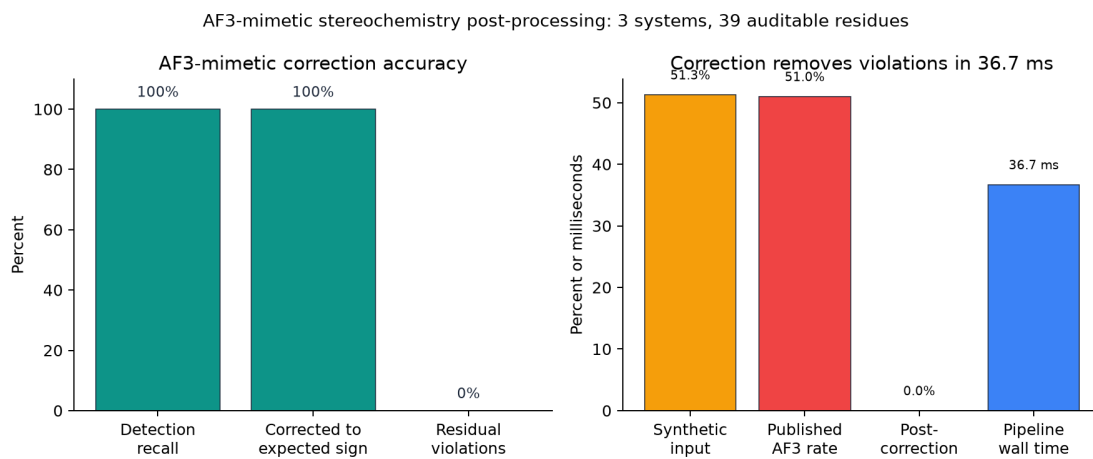


Figure 4: Synthetic AF3-mimetic resource benchmark. ChiralFold correction is a fast stereochemistry post-processing step: 100% detection/correction recall, 0% residual violations, and 0.0367 s total local pipeline time across three Childs-system peptides.